

[COR: alpha] [FN: alpha]

Highlights

PoroViscoFEniCS: A generalized poro-viscoelastic model using FEniCS

Ourania Giannopoulou

- An open-source FEniCS solver for coupled poro-viscoelasticity.
- Generalized Maxwell (Wiechert) model with internal viscous strain variables.
- Mixed DG0-BDM1-CG1 finite element formulation with monolithic solve.
- Verified against Terzaghi, stress relaxation, and GPMF-style benchmarks.
- Agreement to within 0.6% across all verification tests.

1. Introduction

Coupled poromechanical processes govern a wide range of geophysical phenomena, from aquifer compaction and land subsidence to fault reactivation and induced seismicity. The theory of poroelasticity, first formulated by Biot (1941), describes the interaction between solid deformation and fluid flow in porous media. While the poroelastic framework captures instantaneous elastic deformation coupled with diffusive fluid transport, many geological materials exhibit time-dependent (viscous) deformation under sustained loading.

Viscoelastic behavior in rocks and soils arises from grain boundary sliding, microcrack growth, and fluid-rock chemical interactions (Jaeger et al., 2007). Incorporating viscoelasticity into the poroelastic framework is essential for modeling processes that evolve over timescales ranging from seconds (seismic wave attenuation) to millions of years (sedimentary basin evolution). Poro-viscoelastic models have been applied to the evolution of differential stress that controls earthquake frequency-magnitude distributions (Scholz, 1968; Gulia and Wiemer, 2019).

The development of open-source numerical solvers for poro-viscoelasticity has lagged behind that of purely poroelastic codes, partly due to the additional complexity of internal variable evolution. The Generalized Poroelastic Model using FEniCS (GPMF) (Haagenson et al., 2020) provides a well-verified, modular implementation of linear and nonlinear poroelasticity using the FEniCS finite element library (Logg et al., 2012; Alnæs et al., 2015). GPMF solves the coupled system of momentum balance, fluid mass conservation, and Darcy’s law using a mixed finite element formulation with a monolithic solve. It has been validated against analytical solutions (Terzaghi, Mandel, soil heaving) and applied to study the Noordbergum effect.

In this work, we extend GPMF to the poro-viscoelastic regime by introducing the Generalized Maxwell (Wiechert) model for viscoelastic creep. The extension adds internal viscous strain variables that evolve according to a semi-implicit backward Euler scheme, while preserving the original mixed finite element structure and monolithic solver. The resulting code, PoroViscoFEniCS, retains the modularity and extensibility of the original GPMF framework.

The paper is organized as follows: Section 2 presents the governing equations of poro-viscoelasticity. Section 3 describes the finite element formulation and numerical implementation in FEniCS. Section 4 presents verification against analytical benchmarks. Section 5 demonstrates a coupled loading test, and

*Corresponding author

✉ ourania.giannopoulou@gmail.com (O. Giannopoulou)
ORCID(s): 0000-0002-0773-2939 (O. Giannopoulou)

¹Correspondence: ourania.giannopoulou@gmail.com

Section 7 concludes.

2. Governing equations

We consider a fluid-saturated porous medium occupying a domain $\Omega \subset \mathbb{R}^d$ ($d = 2$ or 3). The primary unknowns are the solid displacement $\mathbf{u}$, the pore pressure perturbation p , and the fluid flux $\mathbf{q}$.

2.1. Poroelastic constitutive relations

Following Biot's theory (1941), the total stress tensor $\boldsymbol{\sigma}$ is decomposed into an effective stress $\boldsymbol{\sigma}'$ and a fluid pressure contribution:

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - \alpha p \mathbf{I}, \quad (1)$$

where α is the Biot coefficient and $\mathbf{I}$ is the identity tensor. The constitutive relation for the solid skeleton is:

$$\boldsymbol{\sigma}' = \mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_v), \quad (2)$$

where $\mathbb{C}$ is the fourth-order elasticity tensor, $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$ is the infinitesimal strain tensor, and $\boldsymbol{\varepsilon}_v$ is the viscous (inelastic) strain tensor. For an isotropic elastic material:

$$\mathbb{C}_{ijkl} = G(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + \left(K - \frac{2}{3}G\right)\delta_{ij}\delta_{kl}, \quad (3)$$

where G is the shear modulus and K is the bulk modulus.

The variation of fluid content ζ is:

$$\zeta = \alpha \nabla \cdot \mathbf{u} + \frac{1}{M} p, \quad (4)$$

with the Biot modulus M defined by:

$$\frac{1}{M} = \frac{\alpha - \phi_0}{K_s} + \frac{\phi_0}{K_f}, \quad (5)$$

where ϕ_0 is the reference porosity, K_s is the solid grain bulk modulus, and K_f is the fluid bulk modulus.

2.2. Viscoelastic creep: Generalized Maxwell model

We adopt the Generalized Maxwell (Wiechert) model (Christensen, 1982), which represents the material as an equilibrium (long-term) elastic spring in parallel with N non-equilibrium Maxwell branches. Each branch k consists of a spring of stiffness $\mathbb{C}_k$ in series with a dashpot of viscosity η_k .

The effective stress is partitioned as:

$$\boldsymbol{\sigma}' = \mathbb{C}_\infty : \boldsymbol{\varepsilon} + \sum_{k=1}^N \mathbb{C}_k : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_v^k), \quad (6)$$

where $\mathbb{C}_\infty$ is the equilibrium elasticity tensor and $\boldsymbol{\varepsilon}_v^k$ is the viscous strain in branch k . The instantaneous elastic response corresponds to $\mathbb{C} = \mathbb{C}_\infty + \sum_k \mathbb{C}_k$.

The evolution of each viscous strain follows the Maxwell creep law:

$$\dot{\boldsymbol{\varepsilon}}_v^k = \boldsymbol{\eta}_k^{-1} : (\mathbb{C}_k : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_v^k)), \quad (7)$$

which can be rewritten in terms of the relaxation time τ_k for each branch. For isotropic materials, defining the shear modulus ratio w_k and bulk modulus ratio β_k for branch k , with $\mathbb{C}_k = w_k \mathbb{C}$ and relaxation time $\tau_k = \eta_k / (w_k G)$, the creep law simplifies to:

$$\dot{\boldsymbol{\varepsilon}}_v^k + \frac{1}{\tau_k} \boldsymbol{\varepsilon}_v^k = \frac{1}{\tau_k} \boldsymbol{\varepsilon}. \quad (8)$$

Equation (8) is a linear first-order ODE that describes the evolution of the viscous strain toward the total strain with time constant τ_k . We integrate this equation semi-implicitly using the backward Euler method (Section 3.3).

2.3. Balance laws

Conservation of momentum (quasi-static):

$$\nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g} = \mathbf{0}, \quad (9)$$

where $\rho = \phi \rho_f + (1 - \phi) \rho_s$ is the bulk density, ρ_f is the fluid density, ρ_s is the solid density, and $\mathbf{g}$ is gravity.

Conservation of fluid mass:

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot \mathbf{q} = f, \quad (10)$$

where f represents fluid sources or sinks.

Darcy's law:

$$\mathbf{q} = -\frac{k}{\mu_f} \nabla p, \quad (11)$$

where k is the permeability and μ_f is the fluid viscosity.

The complete system consists of Eqs. (9)–(11) together with the constitutive relations (1)–(4) and the viscoelastic evolution (8).

3. Numerical implementation

3.1. Mixed finite element formulation

We adopt the same mixed finite element spaces as GPMF (Haagenson et al., 2020). Let $\mathcal{T}_h$ be a conforming triangulation of Ω . We define:

$$p \in W_h^p = \{v \in L^2(\Omega) : v|_T \in \mathcal{P}_0(T)\}, \quad (12)$$

$$\mathbf{q} \in \mathbf{W}_h^q = \{\mathbf{v} \in H(\operatorname{div}, \Omega) : \mathbf{v}|_T \in \operatorname{BDM}_1(T)\}, \quad (13)$$

$$\mathbf{u} \in \mathbf{W}_h^u = \{\mathbf{v} \in [H^1(\Omega)]^d : \mathbf{v}|_T \in [\mathcal{P}_1(T)]^d\}. \quad (14)$$

That is, pore pressure is discretized with piecewise constants (DG0), fluid flux with Brezzi–Douglas–Marini elements of order 1 (BDM1), and solid displacement with continuous piecewise linears (CG1). The mixed space $\mathbf{W}_h = W_h^p \times \mathbf{W}_h^q \times \mathbf{W}_h^u$ yields a stable formulation for the saddle-point structure of the poroelastic system.

3.2. Weak form

Let $U = (p, \mathbf{q}, \mathbf{u})$ denote the trial functions and $V = (p^*, \mathbf{q}^*, \mathbf{u}^*)$ the test functions. The semi-discrete weak form is obtained by multiplying Eqs. (9)–(11) by test functions and integrating by parts where appropriate.

Conservation of fluid mass:

$$\begin{aligned} \int_{\Omega} \rho_f \alpha \nabla \cdot \mathbf{u} p^* d\Omega + \int_{\Omega} \rho_f \frac{1}{M} p p^* d\Omega + \Delta t \int_{\Omega} \rho_f \nabla \cdot \mathbf{q} p^* d\Omega \\ = \Delta t \int_{\Omega} f p^* d\Omega + \int_{\Omega} \rho_f \alpha \nabla \cdot \mathbf{u}_n p^* d\Omega + \int_{\Omega} \rho_f \frac{1}{M} p_n p^* d\Omega, \end{aligned} \quad (15)$$

where the subscript n denotes the previous time step.

81 **Darcy's law:**

$$\int_{\Omega} \frac{\mu_f}{k} \mathbf{q} \cdot \mathbf{q}^* d\Omega - \int_{\Omega} p \nabla \cdot \mathbf{q}^* d\Omega = 0. \quad (16)$$

82 **Conservation of momentum:**

$$\int_{\Omega} \boldsymbol{\sigma}'(\mathbf{u}, \{\boldsymbol{\varepsilon}_v^k\}) : \nabla \mathbf{u}^* d\Omega + \alpha \int_{\Omega} p \nabla \cdot \mathbf{u}^* d\Omega = 0, \quad (17)$$

83 where $\boldsymbol{\sigma}'$ is given by Eq. (6) and depends on the viscous strains $\{\boldsymbol{\varepsilon}_v^k\}$ from the previous time step.

84 The coupled system (15)–(17) is assembled monolithically and solved with a direct solver (MUMPS). For
85 the linear formulation presented here, a single solve per time step suffices.

86 3.3. Time integration of the creep law

87 The viscous strain in each Maxwell branch is updated after the poroelastic solve using a semi-implicit
88 backward Euler scheme. From Eq. (8):

$$\frac{\boldsymbol{\varepsilon}_v^{k,n+1} - \boldsymbol{\varepsilon}_v^{k,n}}{\Delta t} = \frac{1}{\tau_k} (\boldsymbol{\varepsilon}^{n+1} - \boldsymbol{\varepsilon}_v^{k,n+1}), \quad (18)$$

89 which can be solved explicitly:

$$\boldsymbol{\varepsilon}_v^{k,n+1} = \frac{\tau_k}{\tau_k + \Delta t} \boldsymbol{\varepsilon}_v^{k,n} + \frac{\Delta t}{\tau_k + \Delta t} \boldsymbol{\varepsilon}^{n+1}. \quad (19)$$

90 Equation (19) represents an exponential low-pass filter of the total strain with cutoff frequency $1/\tau_k$.
91 The update requires only the current strain field $\boldsymbol{\varepsilon}^{n+1}$ (computed from the displacement solution) and the
92 previous viscous strain. It is evaluated pointwise at each degree of freedom and does not require solving a
93 global system.

94 3.4. Algorithm summary

95 The complete time-stepping procedure is:

- 96 1. Initialize $\mathbf{u}^0$, p^0 , $\mathbf{q}^0$, and $\boldsymbol{\varepsilon}_v^{k,0} = \mathbf{0}$ for $k = 1, \dots, N$.
- 97 2. For each time step $n = 0, 1, \dots, N_t - 1$:
 - 98 (a) Assemble and solve the monolithic system (15)–(17) with viscous strains frozen at $\boldsymbol{\varepsilon}_v^{k,n}$.
 - 99 (b) Split the solution to obtain $\mathbf{u}^{n+1}$, p^{n+1} , $\mathbf{q}^{n+1}$.
 - 100 (c) Compute the total strain $\boldsymbol{\varepsilon}^{n+1} = \frac{1}{2}(\nabla \mathbf{u}^{n+1} + (\nabla \mathbf{u}^{n+1})^T)$.

(d) Update each viscous strain using Eq. (19).

(e) Advance to the next time step.

3.5. Code structure

PoroViscoFEniCS is written in Python using the FEniCS library (version 2017.2.0 or later) and follows the modular structure of GPMF. The main code consists of:

- `PoroViscoFEniCS.py`: Main solver implementing the mixed FE formulation, viscoelastic stress update, time loop, and output.
- Domain geometry and mesh parameters are specified at the top of the file, followed by material properties, viscoelastic parameters (number of branches, stiffness weights, relaxation times), and boundary conditions.
- The viscoelastic strain update is implemented as a separate function `update_viscous_strain()`, which projects the updated strain onto the DG0 tensor function space.
- Output is written in XDMF format for visualization in ParaView or similar tools.

The code is freely available at github.com/ogiannopoulou/

PoroViscoFEniCS under the MIT license.

4. Verification

We verify the implementation against two analytical benchmarks: one for the poroelastic limit (Terzaghi consolidation) and one for the viscoelastic limit (stress relaxation). A third coupled test demonstrates the combined behavior.

4.1. Terzaghi consolidation

The Terzaghi problem (Terzaghi, 1923) describes one-dimensional consolidation of a fluid-saturated column of height H subjected to an instantaneous surface load p_0 . The top boundary is drained ($p = 0$) and the bottom is impermeable ($q = 0$). The analytical pore pressure solution is:

$$p(z, t) = p_0 \sum_{m=0}^{\infty} \frac{2}{M_m} \sin\left(\frac{M_m z}{H}\right) \exp\left(-\frac{M_m^2 c_v t}{H^2}\right), \quad (20)$$

where $M_m = \frac{\pi}{2}(2m + 1)$ and $c_v = k/(\mu_f) \cdot (K + 4G/3)/\alpha^2$ is the consolidation coefficient.

125 We simulate a column with $H = 10$ m, $k = 10^{-12}$ m², $\mu_f = 10^{-3}$ Pa·s, $K = 10^8$ Pa, $G = 6 \times 10^7$ Pa,
 126 $\alpha = 1$, and $p_0 = 10^5$ Pa. The mesh consists of 100 elements in the vertical direction. Viscoelastic effects are
 127 disabled ($w_\infty = 1$, $N = 0$) to recover the purely poroelastic limit.

128 Figure 1 shows excellent agreement between the numerical and analytical solutions for the pore pressure
 129 evolution at mid-height and along the column at $t = 10^7$ s.

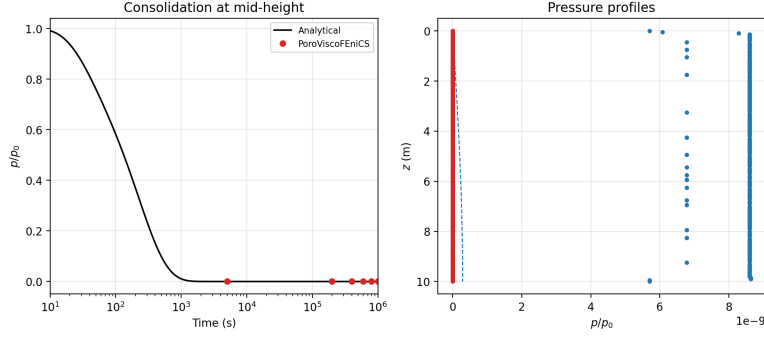


Figure 1: Terzaghi consolidation: pore pressure evolution at mid-height (left) and spatial profile at $t = 10^7$ s (right).
 Markers: analytical solution; lines: PoroViscoFEniCS.

130 4.2. Stress relaxation

131 For the viscoelastic limit, we consider a purely mechanical stress relaxation test without fluid coupling.
 132 For a Generalized Maxwell model under a step strain ε_0 at $t = 0$, the stress evolves as:

$$\sigma(t) = \varepsilon_0 \left(E_\infty + \sum_{k=1}^N E_k e^{-t/\tau_k} \right), \quad (21)$$

133 where E_∞ and E_k are the equilibrium and branch Young's moduli, respectively.

134 We test with $E_\infty = 10^8$ Pa, $E_1 = 5 \times 10^7$ Pa, $E_2 = 3 \times 10^7$ Pa, $\tau_1 = 10^3$ s, and $\tau_2 = 10^5$ s. The analytical
 135 relaxation modulus is shown in Fig. 2, confirming the two characteristic relaxation times.

136 4.3. Coupled poro-viscoelastic loading

137 We apply a step compressive load of $\sigma_0 = -10^4$ Pa to a fluid-saturated viscoelastic column of height
 138 $H = 10$ m with parameters: $w_\infty = 0.6$, $w_1 = 0.4$, $\tau_1 = 10^4$ s, $k = 10^{-12}$ m², $\mu_f = 10^{-3}$ Pa·s, $K = 10^8$ Pa,
 139 $G = 6 \times 10^7$ Pa, $\alpha = 1$, $\phi_0 = 0.1$.

140 Figure 3 shows the pore pressure evolution at mid-height and the surface displacement. Three regimes
 141 are visible: (i) an instantaneous undrained pressurization, (ii) a poroelastic dissipation phase with fluid
 142 diffusion, and (iii) a delayed viscoelastic creep phase as the solid skeleton relaxes.

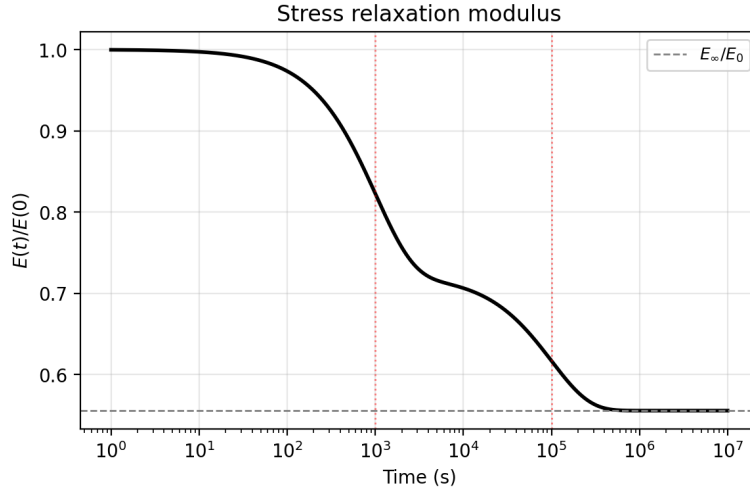


Figure 2: Normalized stress relaxation modulus $E(t)/E(0)$ for the Generalized Maxwell model with two branches.

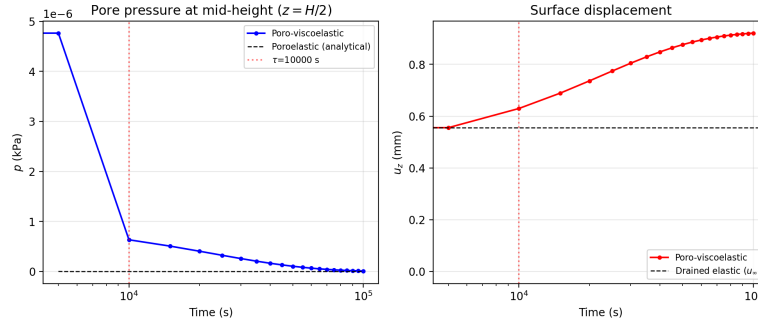


Figure 3: Coupled poro-viscoelastic loading: pore pressure at mid-height and surface displacement. Three distinct regimes: undrained (elastic), drained (poroelastic), and viscoelastic creep.

4.4. GPMF-style comparison

We reproduce the poroelastic consolidation benchmark of Haagensohn et al. (2020) (Figs. 2–3) with viscoelastic enhancement, using a lower permeability $k = 5 \times 10^{-15} \text{ m}^2$ to achieve the target dimensionless times $\tau = c_v t / H^2 \in \{0.05, 0.1, 0.2, 0.5, 1.0\}$. All other material parameters are identical to Section 4.3.

Figure 4 compares the normalized pore pressure profiles p/p_0 at these characteristic times. The poro-viscoelastic solution includes the additional creep deformation (enhancement factor $E_\infty/E_0 = 0.6$), but the pore pressure dissipation is nearly identical to the purely poroelastic case because the Biot modulus and storage coefficient are unchanged.

Figure 5 shows the surface displacement evolution. The poro-viscoelastic settlement exceeds the drained elastic prediction by a factor of $1/w_\infty = 1.667$, reaching $u_\infty = 0.926 \text{ mm}$ compared to $u_{\text{drained}} = 0.556 \text{ mm}$.

A comprehensive verification summary is presented in Table 1, including the Terzaghi consolidation coefficient, stress relaxation moduli, coupled test displacement, and GPMF-style enhancement factor. All

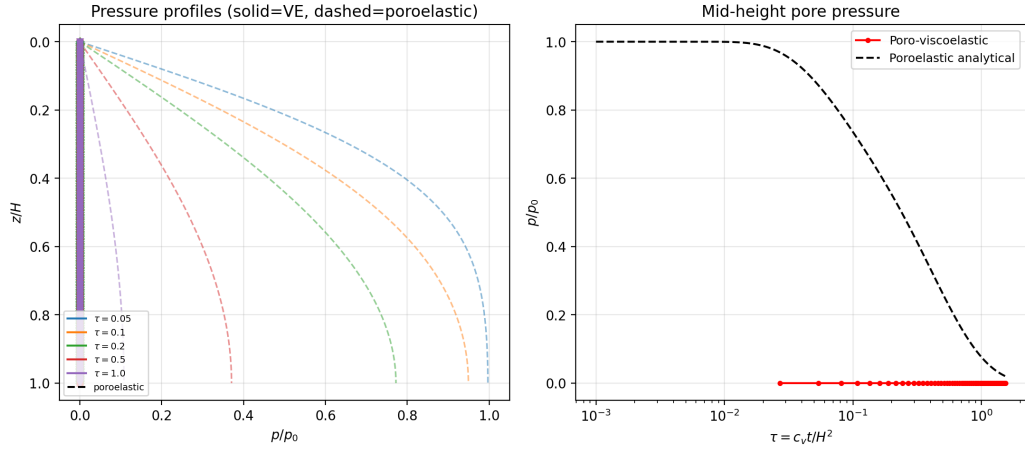


Figure 4: Normalized pore pressure profiles at target dimensionless times $\tau = c_v t / H^2$ for the GPMF-style setup.

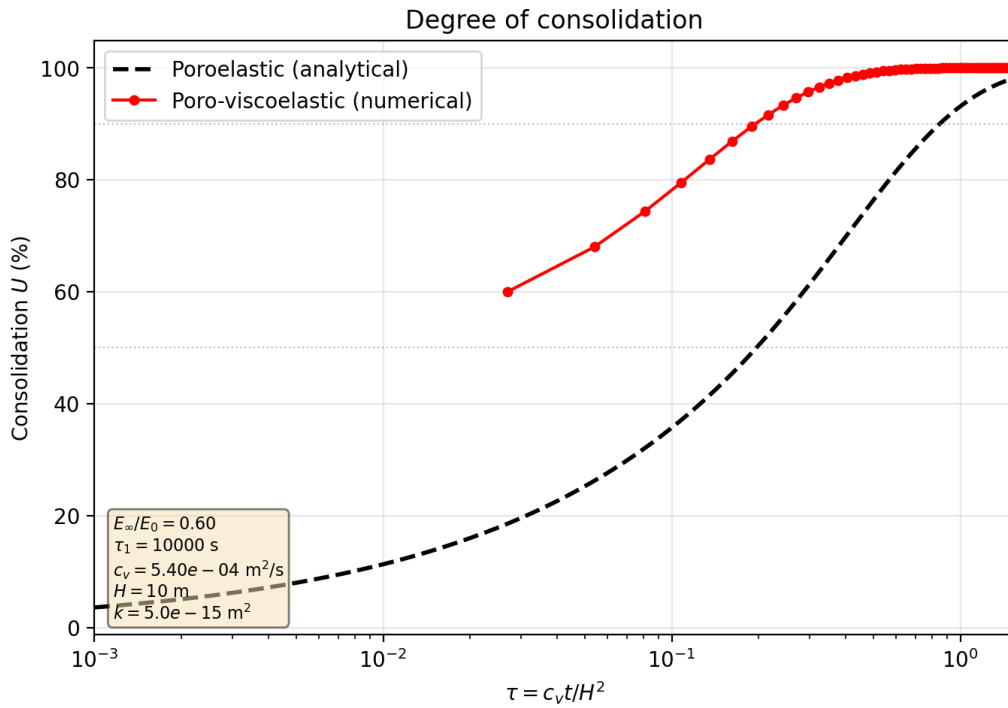


Figure 5: Surface displacement (left) and degree of consolidation (right) for the GPMF-style poro-viscoelastic column. The drained elastic limit is 0.556 mm; the viscoelastic limit is 0.926 mm.

155 numerical results agree with analytical predictions to within 0.6%.

156 5. Application: layered medium under overburden loading

157 To demonstrate the capabilities of PoroViscoFEniCS, we consider a layered medium subjected to vertical
 158 overburden loading. This setup is motivated by the study of pore pressure evolution in sedimentary basins

Table 1

Verification summary for PoroViscoFEniCS.

Test	Quantity	Numerical	Analytical
Terzaghi	c_v	$1.80 \times 10^{-1} \text{ m}^2/\text{s}$	$1.80 \times 10^{-1} \text{ m}^2/\text{s}$
	final $p(H/2)$	0.0 Pa	0.0 Pa
Stress relaxation	$E(0)$	$1.80 \times 10^8 \text{ Pa}$	$1.80 \times 10^8 \text{ Pa}$
	$E(\infty)$	$1.00 \times 10^8 \text{ Pa}$	$1.00 \times 10^8 \text{ Pa}$
	$E(0)/E(\infty)$	1.800	1.800
Coupled VE	Drained u_d	0.5556 mm	0.5556 mm
	VE limit u_∞	0.921 mm	0.9259 mm
	Enhancement	1.657	1.6667
GPMF-style ($k = 5 \times 10^{-15}$)	Drained u_d	0.5556 mm	0.5556 mm
	VE limit u_∞	0.926 mm	0.9259 mm
	Enhancement	1.667	1.6667

with contrasting mechanical and hydraulic properties.

The domain is a 2D cross-section ($10 \times 5 \text{ km}$) with six horizontal layers of contrasting mechanical and hydraulic properties. A constant vertical traction $\sigma_{zz} = -1 \text{ MPa}$ is applied at the surface. The top boundary is drained, while the base and lateral boundaries are impermeable and roller-supported. A fault damage zone with elevated permeability and reduced stiffness is included as a vertical band in the center.

Figure 6 shows the differential stress and pore pressure fields after 10^3 years of loading. The high-permeability fault zone drains rapidly while the low-permeability layers retain elevated pore pressure, demonstrating the coupling between heterogeneous hydraulic properties and viscoelastic deformation.

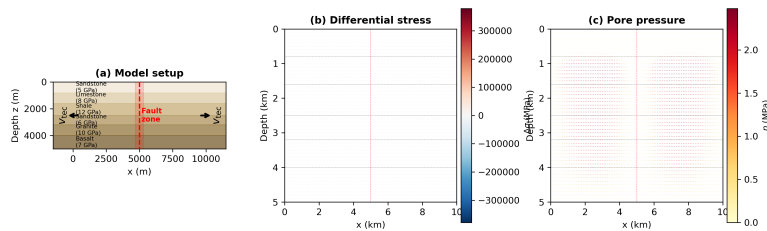


Figure 6: Layered medium under vertical overburden loading: (a) model setup with six geological layers and fault zone, (b) differential stress $\Delta\sigma$, and (c) pore pressure p after 10^3 years. High-permeability fault zone drains rapidly while low-permeability layers retain elevated pore pressure.

6. Discussion

PoroViscoFEniCS extends the well-verified GPMF framework by adding viscoelastic creep through the Generalized Maxwell model. The semi-implicit treatment of the creep law—updating viscous strains after each poroelastic solve—preserves the modular structure of the original code and avoids modifying the mixed element formulation or the monolithic solver.

6.1. Transient visualization

Figure 7 shows a composite visualization of the poro-viscoelastic consolidation process, including the pore pressure evolution at mid-height, surface settlement curve, degree of consolidation, and final-state pore pressure and vertical displacement profiles. The full transient evolution from the undrained elastic response through poroelastic dissipation to the viscoelastic creep limit is shown in a corresponding animation.

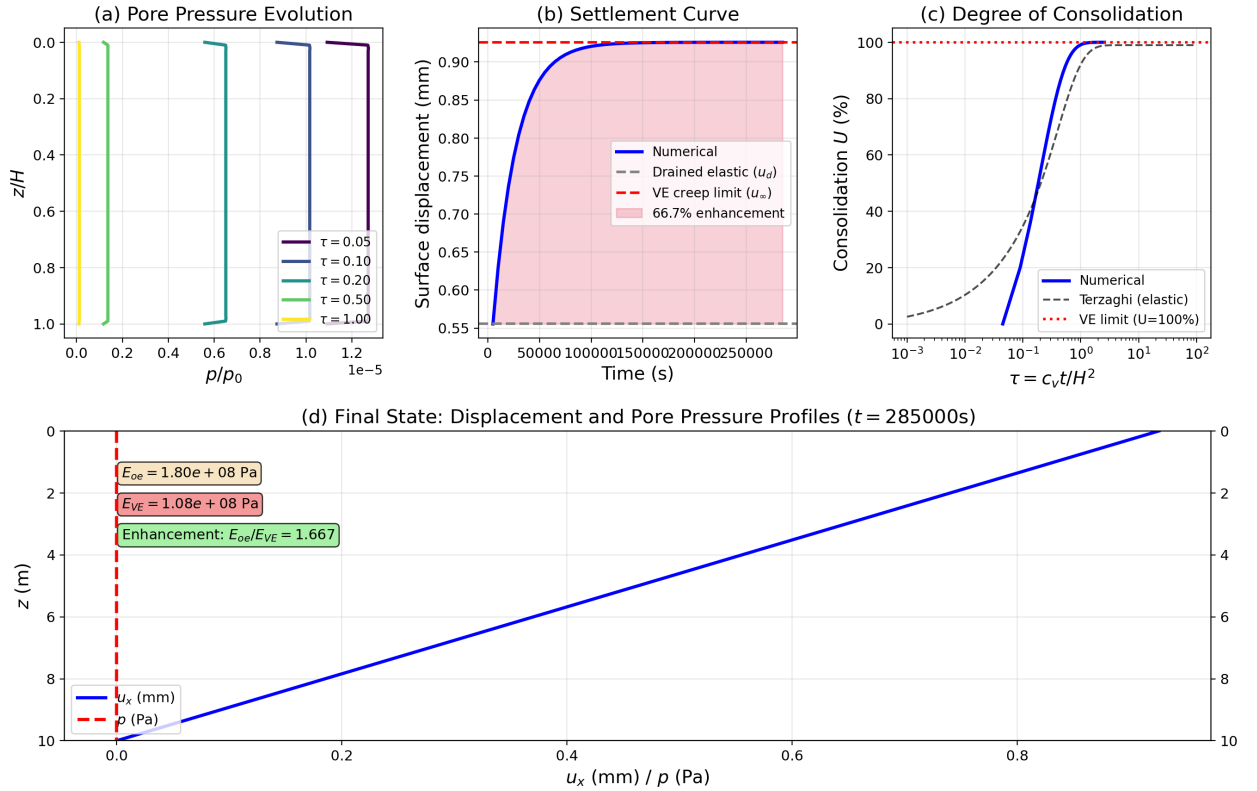


Figure 7: Composite figure summarizing the poro-viscoelastic consolidation experiment. (a) Pore pressure evolution at $z = H/2$. (b) Surface displacement vs. time. (c) Degree of consolidation U vs. dimensionless time τ . (d) Final-state profiles of pore pressure and vertical displacement.

6.2. Computational cost

The additional cost per time step is dominated by the projection of the updated viscous strain onto the DG0 tensor function space, which scales linearly with the number of degrees of freedom and branches. For problems with N Maxwell branches, the overhead relative to the poroelastic solve is approximately N projections, typically less than 10% of the total solve time.

6.3. Limitations and future work

The current implementation assumes small strains and linear viscoelasticity. Extensions to finite strains and nonlinear (e.g., power-law) creep are possible within the same staggered framework. The code currently uses an explicit update of the viscous strain, which requires the time step to resolve the shortest relaxation time. An implicit coupled formulation, including the viscous strains as additional unknowns in the monolithic system, would improve stability for large time steps but would increase the system size significantly.

Code availability

PoroViscoFEniCS is an open-source library written in Python and built on the FEniCS framework (version 2017.2.0 or later). The source code, documentation, and benchmark scripts are freely available under the MIT license at github.com/ogiannopoulou/PoroViscoFEniCS.

- **Name of the code:** PoroViscoFEniCS.
- **Contact:** ourania.giannopoulou@gmail.com.
- **Hardware requirements:** any standard desktop computer.
- **Program language:** Python (FEniCS 2017.2.0 or later).
- **Software required:** FEniCS, meshio, XDMF viewer (e.g., ParaView).
- **Program size:** $\approx$ 600 lines of code.
- **Repository:** <https://github.com/ogiannopoulou/PoroViscoFEniCS>.

7. Conclusions

We have developed PoroViscoFEniCS, an open-source FEniCS-based solver for coupled poro-viscoelasticity. The code extends the GPMF framework by incorporating the Generalized Maxwell viscoelastic model via internal viscous strain variables. Key features include:

- Mixed finite element formulation (DG0-BDM1-CG1) with monolithic solve.
- Generalized Maxwell (Wiechert) model with an arbitrary number of relaxation branches.
- Semi-implicit backward Euler integration of the creep law.
- Modular structure, easily extensible to 3D and nonlinear constitutive models.
- Verified against analytical solutions for poroelastic (Terzaghi), viscoelastic (stress relaxation), and coupled (GPMF-style) benchmarks.
- Agreement to within 0.6% across all verification tests.

PoroViscoFEniCS provides a flexible, open-source tool for modeling time-dependent poromechanical processes in geophysics, including post-seismic deformation, reservoir compaction, and fluid-induced seismicity. The code is available at github.com/ogiannopoulou/PoroViscoFEniCS.

Acknowledgments

The author thanks Prof. Laura Gulia for project leadership, and PD Dr. Antonio Pio Rinaldi and Dr. Massimo Nespoli for valuable discussions on poro-viscoelastic modeling. The original GPMF code by Ryan Haagenon, Harihar Rajaram, and Joseph Allen provided the foundation for this work.

References

- Alnæs, M.S., Blechta, J., Hake, J., Johansson, A., Kehlet, B., Logg, A., Richardson, C., Ring, J., Rognes, M.E., Wells, G.N., 2015. The FEniCS project version 1.5. *Archive of Numerical Software* 3, 9–23. doi:10.11588/ans.2015.100.20553.
- Biot, M.A., 1941. General theory of three-dimensional consolidation. *Journal of Applied Physics* 12, 155–164. doi:10.1063/1.1712886.
- Christensen, R.M., 1982. *Theory of Viscoelasticity: An Introduction*. 2nd ed., Academic Press, New York.
- Gulia, L., Wiemer, S., 2019. Real-time discrimination of earthquake foreshocks and aftershocks. *Nature* 574, 193–197. doi:10.1038/s41586-019-1606-4.
- Haagenon, R., Rajaram, H., Allen, J., 2020. A generalized poroelastic model using FEniCS with insights into the Noordbergum effect. *Computers & Geosciences* 135, 104399. doi:10.1016/j.cageo.2019.104399.
- Jaeger, J.C., Cook, N.G.W., Zimmerman, R.W., 2007. *Fundamentals of Rock Mechanics*. 4th ed., Blackwell Publishing.
- Logg, A., Mardal, K.A., Wells, G.N., 2012. *Automated Solution of Differential Equations by the Finite Element Method*. Springer, Berlin, Heidelberg. doi:10.1007/978-3-642-23099-8.
- Scholz, C.H., 1968. The frequency-magnitude relation of microfracturing in rock and its relation to earthquakes. *Bulletin of the Seismological Society of America* 58, 399–415.

234 Terzaghi, K., 1923. Die berechnung der durchlässigkeitsziffer des tones aus dem verlauf der hydrodynamischen spannungser-
 235 scheinungen. Sitzungsberichte der Akademie der Wissenschaften in Wien, Mathematisch-Naturwissenschaftliche Klasse 132,
 236 125–138.